

Lab 1.1

```
nd6 options=21<PERFORMNUD,AUTO_LINKLOCAL>
[2.8.1-RELEASE][root@pfSense.home.arpal/root]: ping -s 1472 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 1472 data bytes

^Z
Suspended
[2.8.1-RELEASE][root@pfSense.home.arpal/root]: ping -s 1472 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 1472 data bytes

^Z
Suspended
[2.8.1-RELEASE][root@pfSense.home.arpal/root]: ping -s 1300 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 1300 data bytes
1308 bytes from 8.8.8.8: icmp_seq=0 ttl=255 time=222.802 ms
1308 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=33.739 ms
1308 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=37.254 ms
1308 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=38.108 ms
1308 bytes from 8.8.8.8: icmp_seq=4 ttl=255 time=34.318 ms
1308 bytes from 8.8.8.8: icmp_seq=5 ttl=255 time=39.377 ms
1308 bytes from 8.8.8.8: icmp_seq=6 ttl=255 time=35.103 ms
1308 bytes from 8.8.8.8: icmp_seq=7 ttl=255 time=33.416 ms
1308 bytes from 8.8.8.8: icmp_seq=8 ttl=255 time=47.546 ms
1308 bytes from 8.8.8.8: icmp_seq=9 ttl=255 time=34.074 ms

^Z
Suspended
[2.8.1-RELEASE][root@pfSense.home.arpal/root]:
```

Lab 1.2

```
1308 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=33.739 ms
1308 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=37.254 ms
1308 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=38.108 ms
1308 bytes from 8.8.8.8: icmp_seq=4 ttl=255 time=34.318 ms
1308 bytes from 8.8.8.8: icmp_seq=5 ttl=255 time=39.377 ms
1308 bytes from 8.8.8.8: icmp_seq=6 ttl=255 time=35.103 ms
1308 bytes from 8.8.8.8: icmp_seq=7 ttl=255 time=33.416 ms
1308 bytes from 8.8.8.8: icmp_seq=8 ttl=255 time=47.546 ms
1308 bytes from 8.8.8.8: icmp_seq=9 ttl=255 time=34.074 ms

^Z
Suspended
[2.8.1-RELEASE][root@pfSense.home.arpal/root]: ifconfig vtnet0 down
ifconfig: interface vtnet0 does not exist
[2.8.1-RELEASE][root@pfSense.home.arpal/root]: ifconfig em1
em1: flags=1008843<UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST,LOWER_UP> metric 0 mtu
1400
options=48100bb<RXCSUM,TXCSUM,ULAN_MTU,ULAN_HWTAGGING,JUMBO_MTU,ULAN_HWC
SUM,ULAN_HWFILTER,HWSTATS,MEXTPG>
ether 08:00:27:d3:73:bb
inet 192.168.10.1 netmask 0xffffffff broadcast 192.168.10.255
inet6 fe80::a00:27ff:fed3:73bb%em1 prefixlen 64 scopeid 0x2
media: Ethernet autoselect (1000baseT <full-duplex>)
status: active
nd6 options=21<PERFORMNUD,AUTO_LINKLOCAL>
[2.8.1-RELEASE][root@pfSense.home.arpal/root]:
```

Lab 1.3

```
pfsync
pfsync0
[2.8.1-RELEASE][root@pfSense.home.arpal/root]: ifconfig em2
ifconfig: interface em2 does not exist
[2.8.1-RELEASE][root@pfSense.home.arpal/root]: ifconfig opt1 down
ifconfig: interface opt1 does not exist
[2.8.1-RELEASE][root@pfSense.home.arpal/root]: ifconfig em1 down
[2.8.1-RELEASE][root@pfSense.home.arpal/root]: ifconfig em1
em1: flags=1008c02<BROADCAST,DRV_OACTIVE,SIMPLEX,MULTICAST,LOWER_UP> metric 0 mtu 1400
        options=48100bb<RXCSUM,TXCSUM,VLAN_MTU,VLAN_HWTAGGING,JUMBO_MTU,VLAN_HWCSUM,VLAN_HWFILTER,HWSTATS,MEXTPG>
        ether 08:00:27:d3:73:bb
        inet 192.168.10.1 netmask 0xffffffff broadcast 192.168.10.255
        inet6 fe80::a00:27ff:fed3:73bb%em1 prefixlen 64 tentative scopeid 0x2
        media: Ethernet autoselect (1000baseT <full-duplex>)
        status: active
        nd6 options=21<PERFORMNUD,AUTO_LINKLOCAL>
[2.8.1-RELEASE][root@pfSense.home.arpal/root]: tcpdump -i em1
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on em1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^Z
Suspended
[2.8.1-RELEASE][root@pfSense.home.arpal/root]:
```

Lab 2.1

```
@90 pass in quick on em1 inet proto tcp from 192.168.10.100 to 192.168.56.102 flags S/SA keep state (if-bound) label "USER_RULE: NAT " label "id:1765602773" ridentifier 1765602773
  [ Evaluations: 639      Packets: 0      Bytes: 0      States: 0      ]
  [ Inserted: uid 0 pid 0 State Creations: 0      ]
  [ Last Active Time: N/A ]
@91 pass in quick on em1 inet from <LAN__NETWORK:1> to any flags S/SA keep state (if-bound) label "USER_RULE: Default allow LAN to any rule" label "id:0100000101" ridentifier 100000101
  [ Evaluations: 11      Packets: 0      Bytes: 0      States: 0      ]
  [ Inserted: uid 0 pid 0 State Creations: 0      ]
  [ Last Active Time: N/A ]
@92 anchor "tftp-proxy/*" all
  [ Evaluations: 646      Packets: 0      Bytes: 0      States: 0      ]
  [ Inserted: uid 0 pid 0 State Creations: 0      ]
  [ Last Active Time: N/A ]
[2.8.1-RELEASE][root@pfSense.home.arpal/root: /sbin/ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=255 time=182.644 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=34.293 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=36.131 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=37.779 ms
^Z
Suspended
[2.8.1-RELEASE][root@pfSense.home.arpal/root: ]
```

Lab 2.2

```
Dec 13 05:22:12 pfSense filterlog[83352]: 4,,1000000103,em1,match,block,in,4,0x0,,128,4132,0,DF,6,tcp,40,192.168.56.1,192.168.56.102,60232,80,0,FA,3457092322,1590624605,252,,
Dec 13 05:22:13 pfSense filterlog[83352]: 4,,1000000103,em1,match,block,in,4,0x0,,128,4133,0,DF,6,tcp,40,192.168.56.1,192.168.56.102,60232,80,0,FA,3457092322,1590624605,252,,
Dec 13 05:22:13 pfSense filterlog[83352]: 4,,1000000103,em1,match,block,in,4,0x0,,128,4134,0,DF,6,tcp,40,192.168.56.1,192.168.56.102,60232,80,0,FA,3457092322,1590624605,252,,
Dec 13 05:22:14 pfSense filterlog[83352]: 4,,1000000103,em1,match,block,in,4,0x0,,128,4135,0,DF,6,tcp,40,192.168.56.1,192.168.56.102,60232,80,0,FA,3457092322,1590624605,252,,
Dec 13 05:22:17 pfSense filterlog[83352]: 4,,1000000103,em1,match,block,in,4,0x0,,128,4136,0,DF,6,tcp,40,192.168.56.1,192.168.56.102,60232,80,0,FA,3457092322,1590624605,252,,
Dec 13 05:22:17 pfSense filterlog[83352]: 4,,1000000103,em1,match,block,in,4,0x0,,128,4137,0,DF,6,tcp,41,192.168.56.1,192.168.56.102,60232,80,1,A,3457092321:3457092322,1590624605,252,,
Dec 13 05:22:22 pfSense filterlog[83352]: 4,,1000000103,em1,match,block,in,4,0x0,,128,4138,0,DF,6,tcp,40,192.168.56.1,192.168.56.102,60232,80,0,FA,3457092322,1590624605,252,,
Dec 13 05:22:31 pfSense filterlog[83352]: 4,,1000000103,em1,match,block,in,4,0x0,,128,4139,0,DF,6,tcp,40,192.168.56.1,192.168.56.102,60232,80,0,RA,3457092323,1590624605,0,,
```

Lab 3.1

```
8) Shell
Enter an option: 8
[2.8.1-RELEASE][root@pfSense.home.arpal]/root: /sbin/pfctl -sn
no nat proto carp all
nat-anchor "natearly/*" all
nat-anchor "natrules/*" all
nat on em0 inet from 127.0.0.0/8 to any port = isakmp -> 10.0.2.15 static-port
nat on em0 inet from 192.168.10.0/24 to any port = isakmp -> 10.0.2.15 static-po
rt
nat on em0 inet from 192.168.56.0/24 to any port = isakmp -> 10.0.2.15 static-po
rt
nat on em0 inet6 from ::1 to any port = isakmp -> fd17:625c:f037:2:a00:27ff:fe5e
:99fc static-port
nat on em0 inet from 127.0.0.0/8 to any -> 10.0.2.15 port 1024:65535
nat on em0 inet from 192.168.10.0/24 to any -> 10.0.2.15 port 1024:65535
nat on em0 inet from 192.168.56.0/24 to any -> 10.0.2.15 port 1024:65535
nat on em0 inet6 from ::1 to any -> fd17:625c:f037:2:a00:27ff:fe5e:99fc port 102
4:65535
no rdr proto carp all
rdr-anchor "tftp-proxy/*" all
rdr on em1 inet proto tcp from 192.168.10.100 to 192.168.10.200 -> 192.168.56.10
2
[2.8.1-RELEASE][root@pfSense.home.arpal]/root: █
```

Lab3.2

```
nat-anchor "natearly/*" all
nat-anchor "natrules/*" all
nat on em0 inet from 127.0.0.0/8 to any port = isakmp -> 10.0.2.15 static-port
nat on em0 inet from 192.168.10.0/24 to any port = isakmp -> 10.0.2.15 static-po
rt
nat on em0 inet from 192.168.56.0/24 to any port = isakmp -> 10.0.2.15 static-po
rt
nat on em0 inet6 from ::1 to any port = isakmp -> fd17:625c:f037:2:a00:27ff:fe5e
:99fc static-port
nat on em0 inet from 127.0.0.0/8 to any -> 10.0.2.15 port 1024:65535
nat on em0 inet from 192.168.10.0/24 to any -> 10.0.2.15 port 1024:65535
nat on em0 inet from 192.168.56.0/24 to any -> 10.0.2.15 port 1024:65535
nat on em0 inet6 from ::1 to any -> fd17:625c:f037:2:a00:27ff:fe5e:99fc port 102
4:65535
no rdr proto carp all
rdr-anchor "tftp-proxy/*" all
rdr on em1 inet proto tcp from 192.168.10.100 to 192.168.10.200 -> 192.168.56.10
2
[2.8.1-RELEASE][root@pfSense.home.arpal]/root: /sbin/pfctl -sn | grep 192.168.10
nat on em0 inet from 192.168.10.0/24 to any port = isakmp -> 10.0.2.15 static-po
rt
nat on em0 inet from 192.168.10.0/24 to any -> 10.0.2.15 port 1024:65535
rdr on em1 inet proto tcp from 192.168.10.100 to 192.168.10.200 -> 192.168.56.10
2
[2.8.1-RELEASE][root@pfSense.home.arpal]/root: █
```

Lab3.3

```

lo0 udp 127.0.0.1:41221 -> 127.0.0.1:53 SINGLE:NO_TRAFFIC
lo0 udp 127.0.0.1:53 <- 127.0.0.1:41221 NO_TRAFFIC:SINGLE
em0 udp fd17:625c:f037:2:a00:27ff:fe5e:99fc[14955] -> 2001:500:1::53[53] S
INGLE:NO_TRAFFIC
em0 udp fd17:625c:f037:2:a00:27ff:fe5e:99fc[32041] -> 2001:500:2::c[53] SI
NGLE:NO_TRAFFIC
em0 udp 10.0.2.15:57801 -> 192.58.128.30:53 MULTIPLE:SINGLE
lo0 udp ::1[38681] -> ::1[53] SINGLE:NO_TRAFFIC
lo0 udp ::1[53] <- ::1[38681] NO_TRAFFIC:SINGLE
em0 udp 10.0.2.15:64094 -> 8.8.8.8:53 MULTIPLE:SINGLE
em0 udp fd17:625c:f037:2:a00:27ff:fe5e:99fc[24994] -> 2001:503:c27::2:30[53]
SINGLE:NO_TRAFFIC
em0 udp fd17:625c:f037:2:a00:27ff:fe5e:99fc[20538] -> 2001:500:9f::42[53]
SINGLE:NO_TRAFFIC
em0 udp fd17:625c:f037:2:a00:27ff:fe5e:99fc[60576] -> 2001:503:c27::2:30[53]
SINGLE:NO_TRAFFIC
em0 udp fd17:625c:f037:2:a00:27ff:fe5e:99fc[29172] -> 2001:500:9f::42[53]
SINGLE:NO_TRAFFIC
em0 ipv6-icmp fe80::a00:27ff:fe5e:99fc -> fe80::2[135] NO_TRAFFIC:NO_TRAFF
IC
em0 udp 10.0.2.15:29160 -> 192.112.36.4:53 SINGLE:NO_TRAFFIC
em0 udp 10.0.2.15:50822 -> 198.97.190.53:53 SINGLE:NO_TRAFFIC
em0 udp 10.0.2.15:17678 -> 198.41.0.4:53 SINGLE:NO_TRAFFIC
em0 udp 10.0.2.15:42775 -> 192.203.230.10:53 SINGLE:NO_TRAFFIC
[2.8.1-RELEASE][root@pfSense.home.arpa]/root:

```

